CG-CRM — tables and how they connect

Every table also has id (UUID), created_at/updated_at, created_by/updated_by → User, is_active, priority and soft delete (deleted_at), from BaseModel.

RolePermission — accounts
role → Role
module
can_view / create / update / delete

Designation — accounts
name
description
(job title lookup)

State — agency
name, code (2 letters)
used by: licences, contracts, carriers, policies, requests

PolicyType — policies
name
(e.g. MAPD, PDP, Life ...)

Audit notes (*Note tables)
Each record keeps a change log in its own note table, written on every create / update / delete:

UserNote, AgentNote, CarrierNote, AgencyNote, CarrierContractNote, AgencyCarrierContractNote, PasswordNote, PolicyTypeNote, CarrierPolicyNote, CertificationNote, ProductNote

Fields: <record> → parent, kind, changes (JSON diff).

role

Role — accounts
name
description
users ↔ User (M2M)

AgentStateLicense — agents
agent → Agent
state → State
license_number, status
start/end date, life, health

CarrierContract — contracts
agent → Agent
carrier → Carrier
writing_number
appointed_states ↔ State (M2M)

policy_type

CarrierPolicy — policies
carrier → Carrier
policy_type → PolicyType
name
available_states ↔ State (M2M)

roles (M2M)

agent

agent

auto-creates one per line of business

carrier

carrier

User — accounts
email, full_name, phone
designation → Designation
roles ↔ Role (M2M)
agent → Agent (1:1, optional)
is_superuser, is_staff

agent (1:1)

Agent — agents
name, aliases, NPN
email, phone (work + personal)
address, DOB, SSN last 4
join_date, start_date

agent

Certification — policies
agent → Agent
carrier → Carrier
line_of_business
due_date (yearly Sept 15)
file (PDF), is_verified

carrier

Carrier — carriers
name, aliases, link, status
lines_of_business (JSON list)
available_states ↔ State (M2M)

carrier

CarrierStateLicense — carriers
carrier → Carrier
state → State
license_number, status
start/end date, life, health

recipient

agent

agent

carrier

carrier

Notification — notifications
recipient → User
request → Request (optional)
title, body, link
read_at

filing notifies admins

Request — requests
type (HR / merch), status, note
agent → Agent, carrier → Carrier
state → State
product → Product
buyer, size, color, quantity

Password — passwords
carrier → Carrier
agent → Agent OR
agency → Agency
username, portal_password, link

agency

Agency — agency
name, aliases, NPN
email, phone

agency

AgencyCarrierContract — contracts
agency → Agency
carrier → Carrier
contract_number
policy_types ↔ PolicyType (M2M)

product

agency

Product — storefront
name, category, product_type
price, colors, sizes
max_quantity, image_url

AgencyStateLicense — agency
agency → Agency
state → State
license_number, status
start/end date

Legend
- Main records
- State licences
- Lookup lists
- Links between records
- Logins & access
- Requests, orders, alerts
- many → one (crow's foot = many)
- many ↔ many
- automatic action (code writes the other table)

How data moves

From the screen to the database and back, plus the actions that write to other tables on their own.

1. The path of every read and write

Page / component
frontend/app/(dashboard)/<area>/
e.g. agents/[id]/agent-profile.tsx
Shows the data, opens dialogs

request → / ← *JSON*

lib/<entity>.ts
frontend/lib/agents.ts, certifications.ts …
Calls the API, maps API fields
to the shapes the UI uses

request → / ← *JSON*

REST API /api/v1/…
e.g. /agents/, /contracts/,
/certifications/{id}/notes/
JWT sign-in; role permissions

request → / ← *JSON*

Django view + serializer
backend/apps/<app>/apis/<entity>/
views.py, serializers.py
validates, saves, writes notes

request → / ← *JSON*

Model → database table
backend/apps/<app>/models/
soft delete: rows keep
deleted_at instead of vanishing

2. Automatic actions — one save writes to another table

Contract → certifications

Contracts page / agent profile
appointment-dialog.tsx
POST /contracts/create/

→

CarrierContract saved
agent + carrier
contracts/apis/contracts/views.py:96

→

add_contract_certifications()
policies/utils.py:235
one per carrier line of business

→

Certification rows
due next Sept 15
shown on the agent profile

Yearly rollover
(run by hand after each deadline)

manage.py add_contract_
certifications [--due YYYY-MM-DD]
policies/management/commands/

→

Every CarrierContract
loops all agent ↔ carrier
contracts

→

add_contract_certifications()
skips lines already covered
for that deadline

→

Certification rows
next year's cycle added;
undated rows get the due date

Request → notifications

Request dialog / shop
components/request-dialog.tsx
POST /requests/create/

→

Request saved
HR (licence, time off …)
or merch order (Product)

→

notify_admins_of_request()
notifications/services.py:28
one row per admin user

→

Notification rows
bell menu in the dashboard
components/notifications-menu.tsx

Any create / update / delete

Any edit dialog
agents, carriers, contracts,
passwords, policies …

→

View saves the record
takes a snapshot
before and after

→

record_note()
<app>/utils.py
diff_snapshots()

→

***Note row**
shown as the change history
on each record's notes panel

3. Agent login view

Agent signs in (/agent)
User.agent links the login to an Agent
GET /api/v1/auth/agent/
app/agent/page.tsx + lib/agent-home.ts

→

agent_home() accounts/apis/auth/views.py
Gathers in one response:
Agent + AgentStateLicense, CarrierContract,
Certification, Password (for that agent only)

→

Role permissions decide what shows
each section needs agent_view.<section>
view permission (RolePermission)
hidden sections come back empty

Where each table is shown

Frontend pages and components, the lib file they use, and the API path behind it.

Screen	Files	Tables shown	lib → API
Agents list, profile, new, edit	app/(dashboard)/agents/… agent-profile.tsx, agent-form-page.tsx, new/certificate-section.tsx	Agent, AgentStateLicense, CarrierContract, Certification, Password, AgentNote	agents, agent-state-licenses, carrier-contracts, certifications, passwords → /agents/, /contracts/, /certifications/, /passwords/
Certifications (inside agent pages)	app/(dashboard)/certifications/certifications-table.tsx, certification-dialog.tsx	Certification, CertificationNote (and the PDF file)	certifications → /certifications/, /certifications/{id}/file/
Carriers list, profile, new, edit	app/(dashboard)/carriers/… carrier-profile.tsx, carrier-form.tsx, carrier-policy-dialog.tsx	Carrier, CarrierStateLicense, available states, CarrierPolicy, contracts & passwords for the carrier	carriers, carrier-policies, carrier-contracts, passwords → /carriers/, /carrier-policies/
Contracts (by agent / by carrier)	app/(dashboard)/contracts/page.tsx, by-carriers/page.tsx, appointment-dialog.tsx	CarrierContract (agent ↔ carrier, writing number, appointed states)	carrier-contracts → /contracts/
Agency	app/(dashboard)/agency/… agency-profile.tsx, agency-contract-dialog.tsx, license-dialog.tsx	Agency, AgencyStateLicense, AgencyCarrierContract	agency, agency-state-licenses, agency-contracts → /agencies/, /agency-contracts/
Passwords	app/(dashboard)/passwords/… passwords-view.tsx, password-dialog.tsx	Password (agent or agency logins per carrier)	passwords → /passwords/
Policy types	app/(dashboard)/policy-types/…	PolicyType	policy-types → /policy-types/
HR	app/(dashboard)/hr/… hr-view.tsx, hr-calendar.tsx	Request (HR), AgentStateLicense	requests, agent-state-licenses → /requests/
Request dialog (any page)	components/request-dialog.tsx, requests-store.tsx	Request — filing and status changes	requests → /requests/create/, /requests/{id}/
Notification bell	components/notifications-menu.tsx (in dashboard layout)	Notification	notifications → /notifications/, /read/, /read-all/
Storefront & shop	app/(dashboard)/storefront/… storefront-view.tsx, shop/shop-view.tsx, product-dialog.tsx	Product, merch Requests (orders)	storefront, shop → /catalog/, /products/, /requests/merch/
Users & roles	app/(dashboard)/users/…, roles/…	User, Role, RolePermission, UserNote	users, roles → /users/, /roles/
Agent login home	app/agent/page.tsx	Agent, licences, contracts, certifications, passwords	agent-home → /auth/agent/
Global search	app/(dashboard)/layout.tsx	Agents, carriers, requests	search-index (built from agents, carriers, requests)

To trace a field: page → the lib/<entity>.ts it imports (API path + field mapping) → backend/apps/<app>/apis/<entity>/views.py & serializers.py → models/.